MEDIUM

PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

10

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

23, 27, 27, 32, 35, 36, 52

What is the range of the 7 scores shown?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 PROBLEM-SOLVING AND DATA ANALYSIS

At a particular track meet, the ratio of coaches to athletes is 1 to 26. If there are x coaches at the track meet, which of the following expressions represents the number of athletes at the track meet?

A. $\frac{x}{26}$

B. $26x$

C. $x + 26$

D. $\frac{26}{x}$

Q3 PROBLEM-SOLVING AND DATA ANALYSIS

Thomas installed a new stove in his restaurant. At the time of installation, the stove had a value of \$800. Thomas estimates that each year the value of the stove will depreciate by 20% of the previous year's estimated value. What is the estimated value of the stove exactly 2 years after Thomas installed it?

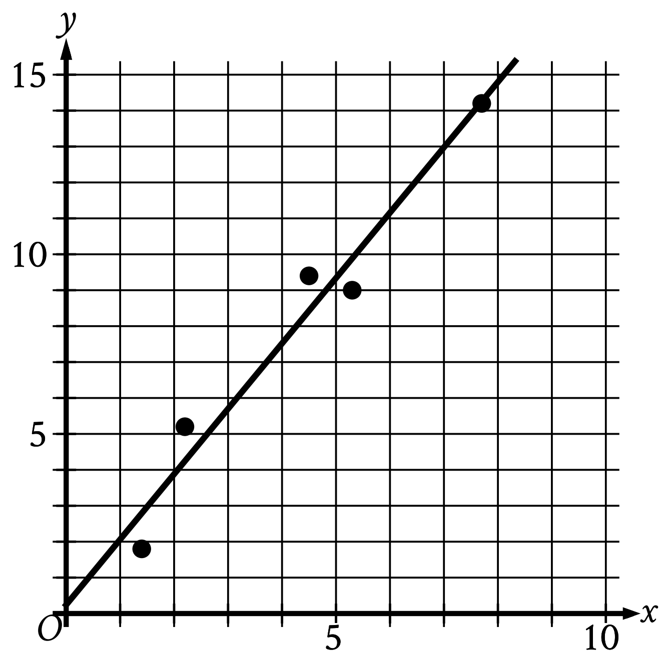
A. \$480

B. \$512

C. \$556

D. \$640

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 0.2)
    - (5 comma 9.3)

Which of the following is closest to the slope of the line of best fit shown?

A. 0.2

B. 0.7

C. 1.8

D. 2.6

Q5 PROBLEM-SOLVING AND DATA ANALYSIS

$$4, 10, 18, 4, 4, 5, 6, 5$$

What is the median of the data set shown?

A. 4

B. 5

C. 7

D. 14

Q6 PROBLEM-SOLVING AND DATA ANALYSIS

An insect moves at a speed of $\frac{3}{20}$ feet per second. What is this speed, in yards per second? $3 \text{ feet} = 1 \text{ yard}$

A. $\frac{1}{20}$

B. $\frac{9}{20}$

C. 6

D. 20

Q7

A customer's monthly water bill was \$75.74. Due to a rate increase, her monthly bill is now \$79.86. To the nearest tenth of a percent, by what percent did the amount of the customer's water bill increase?

A. 4.1%

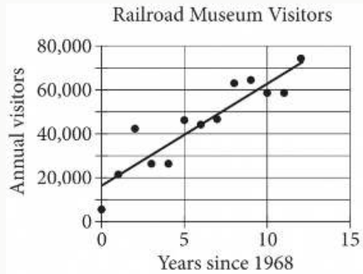
B. 5.1%

C. 5.2%

D. 5.4%

Q8

PROBLEM-SOLVING AND DATA ANALYSIS



The scatterplot above shows the number of visitors to a railroad museum in Pennsylvania each year from 1968 to 1980, where t is the number of years since 1968 and n is the number of visitors. A line of best fit is also shown. Which of the following could be an equation of the line of best fit shown?

A. $n = 16{,}090 + 4{,}680t$

B. $n = 4{,}690 + 16{,}090t$

C. $n = 16{,}090 + 9{,}060t$

D. $n = 9{,}060 + 16{,}090t$

Q9

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

Data value	Frequency
6	3
7	3
8	8
9	8
10	9
11	11
12	9
13	0
14	6

The frequency table summarizes the 57 data values in a data set. What is the maximum data value in the data set?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 PROBLEM-SOLVING AND DATA ANALYSIS

$$d = 55t$$

The equation above can be used to calculate the distance d, in miles, traveled by a car moving at a speed of 55 miles per hour over a period of t hours. For any positive constant k, the distance the car would have traveled after $9k$ hours is how many times the distance the car would have traveled after $3k$ hours?

A. 3

B. 6

C. $3k$

D. $6k$